

1 Introduction

1.1 Effective field theories

TODO:

- Explain low energy (IR) and high energy (UV) theory.
- Mass dimension of operators and increasing precision with Λ .

$$\mathcal{L}_{\text{SMEFT}} = \mathcal{L}_{\text{SM}} + \sum_{d=5}^{N-1} \frac{1}{\Lambda^{d-4}} \sum_n \mathcal{C}_n^{(d)} \mathcal{O}_n^{(d)} + \mathcal{O}\left(\frac{1}{\Lambda^N}\right) \quad (1.1)$$

The Standard Model (SM) Lagrangian $\mathcal{L}_{\text{SM}}$ is supplemented by non-renormalizable operators $\mathcal{O}_n^{(d)}$ of mass dimension d , which incorporate any interactions of SM particles with newly introduced Beyond SM (BSM) particles. These operators can only consist of fields of the low energy theory (SM in this case) and have to obey the symmetries of it, i.e. $\text{SU}(3)_C \otimes \text{SU}(2)_L \otimes \text{U}(1)_Y$ and Lorentz invariance. The so-called Wilson coefficients $\mathcal{C}_n^{(d)}(\mu)$ depend on the renormalization group (RG) scale μ and parameters of the UV theory. It is desirable to parametrize $\mathcal{L}_{\text{SMEFT}}$ with the minimal number of degrees of freedom, i.e. Wilson coefficients which need to be determined by an experiment and still be capable to describe any deviation from the IR theory. Therefore, the Lagrangian is seen as a vector in a vector space, spanned by some operators $\mathcal{O}_n^{(d)}$ with coefficients $\mathcal{C}_n^{(d)}/\Lambda^{d-4}$ and a basis of this vector space of Lagrangians for every mass dimension d has to be found. Such a basis essentially consists of all possible non-redundant operators $\mathcal{O}_n^{(d)}$ for a given level of precision d .

The connection of UV models with low energy observables is done in four steps [s. 1, p. 2–3]:

1. *Basis construction:* A basis consisting of all non-redundant operators build by IR field for each dimension has to be found¹.
2. *Matching:* Then, the UV model has to be matched onto SMEFT at the high energy scale Λ . This matching is done order-by-order in a loop expansion, i.e. $\mathcal{C}_n^{(d)}$ is chosen such that the S matrix element in SMEFT and of the UV model are identical at the renormalization group scale $\mu = \Lambda$ ².
3. *Running:* The RG equation determines the running of the Wilson coefficients $\mathcal{C}_n^{(d)}(\mu)$ and its leading order solution is specified by the anomalous dimension matrix γ_{nm} :

$$\frac{d\mathcal{C}_n^{(d)}}{d \ln \mu} = \sum_m \frac{1}{(4\pi)^2} \gamma_{nm} \mathcal{C}_m^{(d)}. \quad (1.2)$$

After that, the Wilson coefficients can be determined at the scale of the experiment, e.g. the electroweak scale ($\mu = m_W$).

4. *Mapping:* Now, the effective Lagrangian $\mathcal{L}_{\text{SMEFT}}$ can be used to calculate for example observables at the electroweak scale depending only on $\mathcal{C}_n^{(d)}(\mu = m_W)$ and SM parameters.

¹This part is done by Magnus Schaaf and Tim Kempkens.

²This part is done for all gauge theories (?) at one loop order by [2] and has to be converted into our basis. This is done by myself.

2 Construction of the operator basis of

This chapter is focussed on the first step described in the introduction, to find all possible non-redundant operators of the EFT. The generation of them is automatized in the *autoeft* program developed by Magnus Schaaf and Tim Kempkens [3]. In order to create the operators and deal with all redundancies, the program is divided into three parts, dealing with the Lorentz symmetry, the gauge group structure and the flavor indices of the operators. These three parts will be explained in more detail in the following three chapters subsection 2.1, subsection 2.2 and VERWEIS-AUF-FLAVOR.

2.1 Lorentz symmetry

The Lorentz symmetry of the Lagrangian is realized with the help of the proper orthochronous Lorentz group, which is defined by

$$L_+^\uparrow := \{ \Lambda^\mu{}_\nu \in \mathbb{R}^{4 \times 4} | g_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma = g_{\rho\sigma} \wedge \det \Lambda = +1 \wedge \Lambda^0{}_0 \geq 1 \}, \quad (2.1)$$

and a metric signature chosen to be $g_{\mu\nu} = \text{diag}(1, -1, -1, -1)_{\mu\nu}$. Since the group $SL(2, \mathbb{C}) := \{ M \in \mathbb{C}^{2 \times 2} | \det M = +1 \}$ is simply-connected and homeomorphic to $L_+^\uparrow$, i.e. is the universal cover of $L_+^\uparrow$, spinors can be characterized as members of this group.

There exist four different types of Weyl spinors transforming in four different, but pairwise equivalent two-dimensional representations of $SL(2, \mathbb{C})$. For future purpose it is enough to consider only the covariant representation transforming w.r.t. to $A \in SL(2, \mathbb{C})$ and the conjugate contravariant representation transforming with $(A^{-1})^\dagger$. These two representations are inequivalent and the two missing equivalent can be obtained by the relation $\varepsilon A \varepsilon^{-1} = (A^{-1})^T$ with ε given in (2.6)³.

Two spinors $\xi_\alpha(\mathbf{p})$ and $\bar{\chi}^{\dot{\alpha}}(\mathbf{p})$ transforming w.r.t. the mentioned representations transform like:

$$\begin{aligned} \xi_\alpha(\mathbf{p}) &\rightarrow \xi'_\alpha(\mathbf{p}) = A_\alpha{}^\beta \xi_\beta(\Lambda^{-1}(A)\mathbf{p}), \\ \bar{\chi}^{\dot{\alpha}}(\mathbf{p}) &\rightarrow \bar{\chi}'^{\dot{\alpha}}(\mathbf{p}) = \bar{\chi}^{\dot{\beta}}(\Lambda^{-1}(A)\mathbf{p}) (A^{-1})^*{}_{\dot{\beta}}{}^{\dot{\alpha}}, \end{aligned} \quad (2.2)$$

where $\Lambda(A)^\mu{}_\nu := \frac{1}{2} \text{Tr}(\bar{\sigma}^\mu A \sigma_\nu A^\dagger)$ is the A -induced four vector Lorentz transformation. The dot over an index and/or the bar over a function indicates objects transforming w.r.t. A^* instead of A .

σ^μ and $\bar{\sigma}^\mu$ are defined with

$$\boxed{(\sigma^\mu)_{\alpha\dot{\alpha}} := (\mathbb{1}_{\alpha\dot{\alpha}}, (\sigma^i)_{\alpha\dot{\alpha}})^\mu \quad \wedge \quad (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} := (\mathbb{1}^{\dot{\alpha}\alpha}, -(\sigma^i)^{\dot{\alpha}\alpha})^\mu.} \quad (2.3)$$

For our purpose, their most important properties are

$$\boxed{(\sigma^\mu)^{\alpha\dot{\alpha}} (\sigma_\mu)^{\beta\dot{\beta}} = 2\varepsilon^{\alpha\beta} \varepsilon^{\dot{\alpha}\dot{\beta}}} \quad (2.4)$$

and

$$\boxed{(\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} = \varepsilon^{\alpha\beta} \varepsilon^{\dot{\alpha}\dot{\beta}} (\sigma^\mu)_{\beta\dot{\beta}} = (\sigma^\mu)^{\alpha\dot{\alpha}},} \quad (2.5)$$

where the ε -tensor is defined by the Pauli matrix:

$$\varepsilon := +i\sigma^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \varepsilon^{\alpha\beta} := (\varepsilon)^{\alpha\beta}, \quad \varepsilon^{\dot{\alpha}\dot{\beta}} := (\varepsilon)^{\dot{\alpha}\dot{\beta}}. \quad (2.6)$$

³For more detailed derivations and explanations of the following two chapters see [4, p. 30–32] and [5, chap. 34 & 35].

Further identities can be found in appendix A.1.

By introducing hermitian generators of the Lorentz group, the Lie algebra decomposes into 2 independent $\mathfrak{su}(2)$ Lie algebras and hence can be characterized by two half integer numbers (j_l, j_r) . Therefore, $\xi_\alpha \in (\frac{1}{2}, 0)$ can be described as a left-handed and $\bar{\chi}^{\dot{\alpha}} \in (0, \frac{1}{2})$ as a right-handed Weyl spinor.

Standard Model Fields The SM Lagrangian consists only of three types of fields and their derivatives: A scalar (Higgs) field, different kinds of fermions and gauge bosons with their corresponding field strength tensors for each gauge group:

- The Higgs boson H_i and the conjugate Higgs boson $H^{\dagger,i}$ are both in the scalar representation $(0,0)$ and have an additional fundamental (lower) and anti-fundamental (upper) $SU(2)_W$ gauge index.
- The fermions are written as Weyl spinors $\psi_\alpha \in (\frac{1}{2}, 0)$ and $(\psi^\dagger)^{\dot{\alpha}} \in (0, \frac{1}{2})$ and are connected to the four component Dirac spinors by the following relations:

$$\begin{aligned} q_L &= \begin{pmatrix} Q_\alpha \\ 0 \end{pmatrix}, & u_R &= \begin{pmatrix} 0 \\ u_{\mathbb{C}}^{\dagger,\dot{\alpha}} \end{pmatrix}, & d_R &= \begin{pmatrix} 0 \\ d_{\mathbb{C}}^{\dagger,\dot{\alpha}} \end{pmatrix}, & l_L &= \begin{pmatrix} L_\alpha \\ 0 \end{pmatrix}, & e_R &= \begin{pmatrix} 0 \\ e_{\mathbb{C}}^{\dagger,\dot{\alpha}} \end{pmatrix}, \\ \bar{q}_L &= (0, Q_\alpha^\dagger), & \bar{u}_R &= (u_{\mathbb{C}}^\alpha, 0), & \bar{d}_R &= (d_{\mathbb{C}}^\alpha, 0), & \bar{l}_L &= (0, L_\alpha^\dagger), & \bar{e}_R &= (e_{\mathbb{C}}^\alpha, 0). \end{aligned} \quad (2.7)$$

- Field strength tensors are defined in such a way that they are in the $(1,0)$ and $(0,1)$ representation (s. eq. A.31):

$$F_{L\alpha\beta} := \frac{i}{2} F_{\mu\nu} \sigma_{\alpha\beta}^{\mu\nu} \in (1,0) \text{ and } F_R^{\dot{\alpha}\dot{\beta}} := -\frac{i}{2} F^{\mu\nu} \bar{\sigma}_{\mu\nu}^{\dot{\alpha}\dot{\beta}} \in (0,1). \quad (2.8)$$

and are symmetric in their spinor indices. The ordinary field strength tensor can be recovered with:

$$F_{\mu\nu} = -\frac{i}{4} \left(F_R^{\dot{\alpha}\dot{\beta}} (\bar{\sigma}_{\mu\nu})_{\dot{\alpha}\dot{\beta}} - F_{L\alpha\beta} \sigma_{\mu\nu}^{\alpha\beta} \right). \quad (2.9)$$

In both relations we encountered the matrices

$$(\sigma^{\mu\nu})_\alpha{}^\beta := \frac{i}{2} (\sigma^\mu \bar{\sigma}^\nu - \sigma^\nu \bar{\sigma}^\mu)_\alpha{}^\beta \text{ and } (\bar{\sigma}^{\mu\nu})^{\dot{\alpha}}{}_{\dot{\beta}} := \frac{i}{2} (\bar{\sigma}^\mu \sigma^\nu - \bar{\sigma}^\nu \sigma^\mu)^{\dot{\alpha}}{}_{\dot{\beta}}. \quad (2.10)$$

- Finally, the covariant derivative D_μ can be translated in $SL(2, \mathbb{C})$ -notation:

$$D_\alpha^{\dot{\alpha}} := D_\mu (\sigma^\mu)_\alpha{}^{\dot{\alpha}} \in (\frac{1}{2}, \frac{1}{2}) \quad \Leftrightarrow \quad D_\mu = -\frac{1}{2} D_\alpha^{\dot{\alpha}} (\sigma_\mu)_\alpha{}^{\dot{\alpha}}. \quad (2.11)$$

The complete field content of the standard model is then given by:

Fields	$SU(2)_l \otimes SU(2)_r$	h	$SU(3)_C$	$SU(2)_W$	$U(1)_Y$	Flavor	$\dim \Phi_i$
H_i	$(0,0)$	0	1	2	1/2	1	1
$L_{\alpha,i}$	$(\frac{1}{2}, 0)$	-1/2	1	2	-1/2	n_f	3/2
$e_{\mathbb{C}\alpha}$	$(\frac{1}{2}, 0)$	-1/2	1	1	1	n_f	3/2
$Q_{\alpha,a,i}$	$(\frac{1}{2}, 0)$	-1/2	3	2	1/6	n_f	3/2
$u_{\mathbb{C}\alpha}^a$	$(\frac{1}{2}, 0)$	-1/2	$\bar{3}$	1	-2/3	n_f	3/2
$d_{\mathbb{C}\alpha}^a$	$(\frac{1}{2}, 0)$	-1/2	$\bar{3}$	1	1/3	n_f	3/2
$G_{L\alpha\beta}^A$	$(1,0)$	-1	8	1	0	1	2
$W_{L\alpha\beta}^I$	$(1,0)$	-1	1	3	0	1	2
$B_{L\alpha\beta}$	$(1,0)$	-1	1	1	0	1	2

Table 1: The helicity $h = j_r - j_l$ is well-defined, since we are only considering the unbroken phase of SMEFT. This further means that all particles are massless and have a mass dimension of $\dim \Phi_i = 1 + |h_i|$.

Each structure consists of several fields and their derivatives, which motivates the definition of the following building blocks:

$$D^{n_D} \Phi := (D^{r-|h|} \Phi)_{\alpha_1 \dots \alpha_{r-h}}^{\dot{\alpha}_1 \dots \dot{\alpha}_{r+h}} = \begin{cases} D_{\alpha_1 \dots \alpha_{r+h}}^{\dot{\alpha}_1 \dots \dot{\alpha}_{r+h}} \Phi_{\alpha_{r+h+1} \dots \alpha_{r-h}}, & h < 0 \\ D_{\alpha_1 \dots \alpha_{r-h}}^{\dot{\alpha}_1 \dots \dot{\alpha}_{r-h}} \Phi_{\dot{\alpha}_{r-h+1} \dots \dot{\alpha}_{r+h}}, & h > 0 \end{cases}, \quad (2.12)$$

where the helicity h of each particle is given by $h = j_r - j_l$ (cf. Table 1) and $r := n_D + |h|$ is half the number of indices on a building block. Operators can now be constructed from these building blocks in combinations which form a singlet under all symmetry groups of the Lagrangian.

Insert example in this place.

Since we are only concerned with the Lorentz symmetry in this chapter, we construct the Lorentz singlet

$$M_{\{g\},\{h\}}^{\{f\}} = \left[T_{\text{Lorentz},\{\dot{\alpha}\}}^{\{\alpha\}} \prod_{i=1}^N (D^{r_i-|h_i|} \Phi)_{\alpha_1 \dots \alpha_{r_i-h_i}}^{\dot{\alpha}_1 \dots \dot{\alpha}_{r_i+h_i}} \right]_{\{g\},\{h\}}^{\{f\}} \quad (2.13)$$

with the help of invariant coefficient tensors $T_{\text{Lorentz},\{\dot{\alpha}\}}^{\{\alpha\}}$. These tensors are constructed from the presented invariant tensors $(\sigma^{\mu\nu})_{\alpha\beta}$, $(\bar{\sigma}^{\mu\nu})^{\dot{\alpha}\dot{\beta}}$, $(\sigma^\mu)_{\alpha\dot{\alpha}}$, $(\bar{\sigma}^\mu)^{\alpha\dot{\alpha}}$, $\varepsilon^{\alpha\beta}$ and $\varepsilon_{\dot{\alpha}\dot{\beta}}$ w.r.t the Lorentz group. All open Lorentz indices are contracted such, that only $\text{SL}(2, \mathbb{C})$ indices and tensors $\varepsilon^{\alpha\beta}$ and $\varepsilon_{\dot{\alpha}\dot{\beta}}$ occur in $T_{\text{Lorentz},\{\dot{\alpha}\}}^{\{\alpha\}}$. The indices $\{g\}$ and $\{h\}$ are group indices of $\text{SU}(3)_C$ and $\text{SU}(2)_L$ respectively, while $\{f\}$ are flavor indices. The regarding further construction of gauge singlet is examined in the next chapter.

For now, let's consider first of all possible redundancies which may occur and present an algorithm which is capable of constructing Lorentz singlets without those.

TODO:

- Describe all possible redundancies and focus on the parts which have to be tackled later in the projection of the BSUOLEA. This are especially the $[D_\mu, D_\nu]$ and EOM redundancies.
- Construction of Lorentz singlet

2.2 Gauge symmetry

TODO

2.3 Flavor structure

TODO

3 Projection

Insert Motivation for projection.

The goal of this section is to examine some examples of Benjamin Summ's output and convert them in our Notation and Operator Basis. In the beginning, the operators are converted to the correct notation, where

- no fields are directly contracted with other fields,
- only fundamental indices occur,
- "smallest" index, w.r.t. the number on the index should be the first one in each ε ,
- fields are sorted with increasing helicity.

This rewriting allows to write each operator as linear combination $\mathcal{O} = T^{abcdef} \Theta_{abcdef}$.

3.1 Real singlet scalar extension of the SM (SSM)

The electroweak (EW) part of the SM is given by the Lagrangian:

$$\begin{aligned} \mathcal{L}_{\text{SM}} = & (D_\mu H)^\dagger (D^\mu H) + \mu_h^2 H^\dagger H - \frac{1}{2} \lambda_h (H^\dagger H)^2 - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} - \frac{1}{4} W_{\mu\nu}^a W^{a\mu\nu} \\ & + \sum_{f=q,u,d,\ell,e} \bar{f} i \not{D} f - \left(y_u \bar{q} \tilde{H} u + y_d \bar{q} H d + y_e \bar{\ell} H e + \text{h.c.} \right), \end{aligned} \quad (3.1)$$

where H denotes the SM Higgs doublet and $\tilde{H}_i := (i\sigma^2 H^*)_i = -\varepsilon_{ij} (H_j)^*$ denotes the charge-conjugated Higgs field. Note further that right-handed neutrino fields are not considered in this investigation and the sum over flavour indices is implicit. The covariant derivative in the only EW case (QCD is ignored, since it is irrelevant for the considered matching) is specified as

$$D_\mu = \partial_\mu - ig_1 Y B_\mu - ig_2 \frac{\sigma^a}{2} W_\mu^a. \quad (3.2)$$

g_1 and B_μ are the $U(1)_Y$, g_2 and W_μ^a are the $SU(2)_L$ gauge coupling and gauge fields. The Yukawa couplings are y_u , y_d and y_e . q and ℓ denote the left-handed quark and lepton doublets, whereas u , d and e are the right-handed fermion singlets. The Lagrangian of the real scalar singlet is restricted by Lorentz and gauge invariance and is given by

$$\mathcal{L}_\phi = \frac{1}{2} (\partial_\mu \phi)^2 - \frac{1}{2} M^2 \phi^2 - A H^\dagger H \phi - \frac{1}{2} \kappa H^\dagger H \phi^2 - \frac{1}{3!} \mu \phi^3 - \frac{1}{4!} \lambda \phi^4. \quad (3.3)$$

The parameters which don't have mass dimension zero are parametrized by the scalar field mass M :

$$A = aM, \quad \mu = \hat{\mu}M, \quad \mu_h = \hat{\mu}_h M, \quad \mu_M = \hat{\mu}_M M. \quad (3.4)$$

In the BSUOLEA output the constants are specified as follows:

$$d = \text{d}, \quad \varepsilon = \text{eps}, \quad \mu_h = \text{Mu}, \quad \lambda_h = \text{lambdah}, \quad g_1 = \text{g1}, \quad g_2 = \text{g2}, \quad g_3 = \text{g3}, \quad (3.5)$$

$$M = \text{Ms}, \quad A = \text{At}, \quad \kappa = \text{kappa}, \quad \mu = \text{mu}, \quad \lambda_\phi = \text{lambda phi}, \quad \mu_M = \text{[Mu]}, \quad (3.6)$$

where ε denote the epsilon of dimensional regularization (DREG) in d dimensions, i.e. $d = 4 - \varepsilon$. The python program leaves most of the names untouched, but $\mu_M = \text{muM}$ has to be renamed for FORM compatibility. Moreover, the logarithm $\ln \frac{M_s^2}{\mu_M^2} = [2\text{L}[\text{Ms}, \text{muM}]]$ is rewritten.

⁴This is the at least the definition of the DREG- ε in [2, p. 57]. Note, that the DREG- ε in [7, p. 4] is defined with $d = 4 - 2\varepsilon$.

3.2 Examples from Benjamin Summ's direct output

Check examples again and add also examples for IBP relations and the projection procedure of the gauge groups.

Let's consider the 58th Term in Benjamin's output and (A.35), (3.30) and (2.7), where $l_L := \begin{pmatrix} (L)_\alpha \\ 0 \end{pmatrix}$:

$$\begin{aligned}
H^{\dagger,j} H_i (\not{D}(\bar{l}_L)_{i,r}) (y_e y_e^\dagger)_{rt} (l_L)_{j,t} &= H^{\dagger,j} H_i D_\mu (\bar{l}_L)_i \gamma^\mu (y_e y_e^\dagger) (l_L)_j \\
&= \varepsilon^{jk} H_k^\dagger H_i (D_\mu (L^\dagger)_{i,\dot{\alpha}} (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} (y_e y_e^\dagger) L_{j\alpha} \\
&= \varepsilon^{\alpha\beta} \varepsilon_{\dot{\alpha}\dot{\beta}} \varepsilon^{jk} H_k^\dagger H_i (D (L^\dagger)_i)^{\dot{\alpha}\dot{\beta}} (y_e y_e^\dagger) L_{j\alpha} \\
&= \varepsilon^{jk} \varepsilon^{\alpha\beta} \varepsilon_{\dot{\alpha}\dot{\beta}} (y_e y_e^\dagger)_{rt} H_k^\dagger H_i (D (L^\dagger)_i)^{\dot{\alpha}\dot{\beta}} L_{j\alpha}^t
\end{aligned} \tag{3.7}$$

From step two to three (2.4) is used. In a next step, the 35th term is inspected:

$$\begin{aligned}
(T^I T^J)_i^j H^{\dagger,i} H_j V_{\mu\nu}^I V^{J,\mu\nu} &= -\frac{1}{16} \left(\varepsilon^{\alpha_1\beta_2} \varepsilon^{\alpha_2\beta_1} + \varepsilon^{\alpha_1\alpha_2} \varepsilon^{\beta_2\beta_1} \right) (T^I T^J)_i^j \varepsilon^{ik} H_k H_j V_{L\alpha_1\beta_1}^I V_{L\alpha_2\beta_2}^J \\
&\quad - \frac{1}{16} \left(\varepsilon_{\dot{\alpha}_1\dot{\beta}_2} \varepsilon_{\dot{\alpha}_2\dot{\beta}_1} + \varepsilon_{\dot{\alpha}_1\dot{\alpha}_2} \varepsilon_{\dot{\beta}_2\dot{\beta}_1} \right) (T^I T^J)_i^j \varepsilon^{ik} H_k H_j V_{R,\dot{\alpha}_1\dot{\beta}_1}^I V_{R,\dot{\alpha}_2\dot{\beta}_2}^J,
\end{aligned} \tag{3.8}$$

where we used to write the Field strength tensors in the chiral notation. Next, the SU(2) part is rewritten and fully expressed by fundamental indices, with the help of (3.29):

$$\begin{aligned}
\varepsilon^{ik} (T^I T^J)_i^j V^I V^J &= -\varepsilon^{ki} \left((T^I)_{ik} (T^I)^l{}_m \right) \left((T^J)^{kj} (T^J)^p{}_q \right) \varepsilon^{mn} \varepsilon^{qr} V_{(ln)} V_{(pr)} \\
&\quad \vdots \\
&= -\varepsilon^{ki} \varepsilon_i{}^n \varepsilon^{lr} \delta^{pj} V_{(ln)} V_{(pr)} \\
&= -\varepsilon^{ki} \varepsilon^{in} \varepsilon^{lr} \varepsilon^{pq} \varepsilon^{qj} V_{(ln)} V_{(pr)}.
\end{aligned} \tag{3.9}$$

This yields for the whole expression

$$\begin{aligned}
(T^I T^J)_i^j H^{\dagger,i} H_j V_{\mu\nu}^I V^{J,\mu\nu} &= -\frac{1}{16} \left(\varepsilon^{\alpha_1\beta_2} \varepsilon^{\beta_1\alpha_2} + \varepsilon^{\alpha_1\alpha_2} \varepsilon^{\beta_1\beta_2} \right) \varepsilon^{ip} \varepsilon^{pl} \varepsilon^{kn} \varepsilon^{mq} \varepsilon^{qj} H_i^\dagger H_j V_{L\alpha_1\beta_1,(kl)} V_{L\alpha_2\beta_2,(mn)} \\
&\quad - \frac{1}{16} \left(\varepsilon_{\dot{\alpha}_1\dot{\beta}_2} \varepsilon_{\dot{\beta}_1\dot{\alpha}_2} + \varepsilon_{\dot{\alpha}_1\dot{\alpha}_2} \varepsilon_{\dot{\beta}_1\dot{\beta}_2} \right) \varepsilon^{ip} \varepsilon^{pl} \varepsilon^{kn} \varepsilon^{mq} \varepsilon^{qj} H_i^\dagger H_j V_{R,\dot{\alpha}_1\dot{\beta}_1,(kl)} V_{R,\dot{\alpha}_2\dot{\beta}_2,(mn)}.
\end{aligned} \tag{3.10}$$

For term 103 one get's:

$$\begin{aligned}
H_i H_k H^{\dagger,j} \varepsilon_{ik} \varepsilon_{lj} (y_u y_u^\dagger y_d)_{tr} (\bar{q}_L)_{l,t}^a (d_R)_r^a \\
&= \varepsilon_{ik} \varepsilon_{lj} \varepsilon_{jm} (y_u y_u^\dagger y_d)_{tr} H_i H_k H_m^\dagger (0, (Q^\dagger)_{l,t,\dot{\alpha}}^a) \cdot \begin{pmatrix} 0 \\ (d_\mathbb{C}^\dagger)_r^{a,\dot{\alpha}} \end{pmatrix} \\
&= \frac{1}{4} (\delta^{bd} \delta^{ce} - \delta^{be} \delta^{cd}) \varepsilon_{ik} \varepsilon_{lj} \varepsilon_{jm} (y_u y_u^\dagger y_d)_{tr} \varepsilon_{\dot{\alpha}\dot{\beta}} H_i H_k H_m^\dagger (Q^\dagger)_{l,t}^{bc,\dot{\beta}} (d_\mathbb{C}^\dagger)_r^{de,\dot{\alpha}}
\end{aligned} \tag{3.11}$$

From here on the general procedure should be explained.

3.3 Lorentz group

Unfortunately, due to the structure of the Lorentz group all known redundancy relations have to be implemented manually. These relations are:

1. Derivatives should only act on the first field. If this isn't the case integration by parts and the Leibniz rule are used to rewrite them.

$$\left(D^{r_1-|h_1|}\Phi_1\right)\dots\simeq-\Phi_1\left(D^{r_1-|h_1|}\dots\right), \quad (3.12)$$

This has to be done, because otherwise the corresponding YT of the auxiliary $SU(N)$ group aren't SSYT. Further this is just an incorporation of momentum conservation:

$$\begin{aligned} \sum_{k=1}^N k_k^\mu &= 0 \wedge k_k^2 = 0 \\ \Rightarrow \sum_{k=i}^N \langle pk_i \rangle [k_i q] &= 0 \Leftrightarrow \sum_{k=1}^N \varepsilon_{\dot{\alpha}_k \dot{\alpha}_i} \varepsilon^{\alpha_k \alpha_j} = 0 \Leftrightarrow \sum_{k=1}^N \overline{\left[\frac{i}{k}\right]} \otimes \left[\frac{k}{j}\right] = 0 \end{aligned} \quad (3.13)$$

This can be rewritten in terms of Young Diagrams:

$$\begin{aligned} \overline{\left[\frac{1}{i}\right]} \otimes \left[\frac{1}{j}\right] D_{\alpha_1}^{\dot{\alpha}_1} \Phi^{(1)} \dots \Phi^{(i), \dot{\alpha}_i} \dots \Phi_{\alpha_j}^{(j)} \dots &= - \sum_{\substack{k=2, \\ k \notin \{i, j\}}}^N \overline{\left[\frac{i}{k}\right]} \otimes \left[\frac{k}{j}\right] \Phi^{(1)} \dots D_{\alpha_k}^{\dot{\alpha}_k} \Phi^{(k)} \dots \Phi^{(i), \dot{\alpha}_i} \dots \Phi_{\alpha_j}^{(j)} \dots \\ &\quad - \overline{\left[\frac{\dot{\alpha}_i}{\beta_i}\right]} \otimes \left[\frac{\alpha_i}{\alpha_j}\right] \Phi^{(1)} \dots D_{\alpha_i}^{\dot{\alpha}_i} \Phi^{(i), \dot{\beta}_i} \dots \Phi_{\alpha_j}^{(j)} \dots \\ &\quad - \overline{\left[\frac{\dot{\alpha}_j}{\beta_j}\right]} \otimes \left[\frac{\alpha_j}{\beta_i}\right] \Phi^{(1)} \dots \Phi^{(i), \dot{\alpha}_i} \dots D_{\alpha_j}^{\dot{\alpha}_j} \Phi_{\beta_j}^{(j)} \dots \\ \Leftrightarrow \overline{\left[\frac{1}{i}\right]} \otimes \left[\frac{1}{j}\right] &= - \sum_{\substack{k=2, \\ k \notin \{i, j\}}}^N \overline{\left[\frac{i}{k}\right]} \otimes \left[\frac{k}{j}\right] - \overline{\left[\frac{\dot{\alpha}_i}{\beta_i}\right]} \otimes \left[\frac{\alpha_i}{\alpha_j}\right] \{\dot{\alpha}_i \rightarrow \dot{\beta}_i\} - \overline{\left[\frac{\dot{\alpha}_j}{\beta_j}\right]} \otimes \left[\frac{\alpha_j}{\beta_i}\right] \{\alpha_j \rightarrow \beta_j\} \end{aligned} \quad (3.14)$$

2. Spinor indices of a derivative on the second contracted with indices of the first building block should be rewritten:

$$\Phi_{\alpha}^{(1)} \left(D_{\alpha}^{\alpha} D^{n_2-1} \Phi^{(2)}\right) \dots \simeq -\Phi_{\alpha}^{(1)} \left(D^{n_2-1} \Phi^{(2)}\right) \dots (D_{\alpha}^{\alpha} \dots), \quad (3.15)$$

This can be rewritten in terms of Young Diagrams:

$$\begin{aligned} \overline{\left[\frac{2}{i}\right]} \otimes \left[\frac{1}{2}\right] \Phi_{\alpha_1}^{(1)} D_{\alpha_2}^{\dot{\alpha}_2} \Phi^{(2)} \dots \Phi^{(i), \dot{\alpha}_i} \dots &= - \sum_{k=2, k \neq i}^N \overline{\left[\frac{k}{i}\right]} \otimes \left[\frac{1}{k}\right] \Phi_{\alpha_1}^{(1)} \Phi^{(2)} \dots D_{\alpha_k}^{\dot{\alpha}_k} \Phi^{(k)} \dots \Phi^{(i), \dot{\alpha}_i} \dots \\ &\quad + \overline{\left[\frac{\dot{\alpha}_1}{\alpha_i}\right]} \otimes \left[\frac{\alpha_1}{\beta_1}\right] D_{\alpha_1}^{\dot{\alpha}_1} \Phi_{\beta_1}^{(1)} \Phi^{(2)} \dots \Phi^{(i), \dot{\alpha}_i} \dots \\ &\quad - \overline{\left[\frac{\dot{\alpha}_i}{\beta_i}\right]} \otimes \left[\frac{\alpha_1}{\alpha_i}\right] \Phi_{\alpha_1}^{(1)} \Phi^{(2)} \dots D_{\alpha_i}^{\dot{\alpha}_i} \Phi^{(i), \dot{\beta}_i} \dots \\ \Leftrightarrow \overline{\left[\frac{2}{i}\right]} \otimes \left[\frac{1}{2}\right] &= - \sum_{k=2, k \neq i}^N \overline{\left[\frac{k}{i}\right]} \otimes \left[\frac{1}{k}\right] + \overline{\left[\frac{\dot{\alpha}_1}{\alpha_i}\right]} \otimes \left[\frac{\alpha_1}{\beta_1}\right] \{\alpha_1 \rightarrow \beta_1\} - \overline{\left[\frac{\dot{\alpha}_i}{\beta_i}\right]} \otimes \left[\frac{\alpha_1}{\alpha_i}\right] \{\dot{\alpha}_i \rightarrow \dot{\beta}_i\} \end{aligned} \quad (3.16)$$

There exists of course the same relation with changed indices:

$$\Phi_{\alpha}^{(1), \dot{\alpha}} \left(D_{\alpha}^{\alpha} D^{n_2-1} \Phi^{(2)}\right) \dots \simeq -\Phi_{\alpha}^{(1), \dot{\alpha}} \left(D^{n_2-1} \Phi^{(2)}\right) \dots (D_{\alpha}^{\alpha} \dots), \quad (3.17)$$

This can be rewritten in terms of Young Diagrams:

$$\begin{aligned}
\overline{\begin{array}{|c|} \hline 1 \\ \hline 2 \end{array}} \otimes \begin{array}{|c|} \hline 2 \\ \hline i \end{array} \Phi^{(1),\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} \Phi^{(2)} \dots \Phi_{\alpha_i}^{(i)} \dots = - \sum_{k=2, k \neq i}^N \overline{\begin{array}{|c|} \hline 1 \\ \hline k \end{array}} \otimes \begin{array}{|c|} \hline k \\ \hline i \end{array} \Phi^{(1),\dot{\alpha}_1} \Phi^{(2)} \dots D_{\alpha_k}^{\dot{\alpha}_k} \Phi^{(k)} \dots \Phi_{\alpha_i}^{(i)} \dots \\
+ \overline{\begin{array}{|c|} \hline \dot{\alpha}_1 \\ \hline \beta_1 \end{array}} \otimes \begin{array}{|c|} \hline \alpha_1 \\ \hline \alpha_i \end{array} D_{\alpha_1}^{\dot{\alpha}_1} \Phi^{(1),\dot{\beta}_1} \Phi^{(2)} \dots \Phi_{\alpha_i}^{(i)} \dots \\
- \overline{\begin{array}{|c|} \hline \dot{\alpha}_1 \\ \hline \dot{\alpha}_i \end{array}} \otimes \begin{array}{|c|} \hline \alpha_i \\ \hline \beta_i \end{array} \Phi^{(1),\dot{\alpha}_1} \Phi^{(2)} \dots D_{\alpha_i}^{\dot{\alpha}_i} \Phi_{\beta_i}^{(i)} \dots \\
\Leftrightarrow \overline{\begin{array}{|c|} \hline 1 \\ \hline 2 \end{array}} \otimes \begin{array}{|c|} \hline 2 \\ \hline i \end{array} = - \sum_{k=2, k \neq i}^N \overline{\begin{array}{|c|} \hline 1 \\ \hline k \end{array}} \otimes \begin{array}{|c|} \hline k \\ \hline i \end{array} + \overline{\begin{array}{|c|} \hline \dot{\alpha}_1 \\ \hline \beta_1 \end{array}} \otimes \begin{array}{|c|} \hline \alpha_1 \\ \hline \alpha_i \end{array} \{\dot{\alpha}_1 \rightarrow \dot{\beta}_1\} - \overline{\begin{array}{|c|} \hline \dot{\alpha}_1 \\ \hline \dot{\alpha}_i \end{array}} \otimes \begin{array}{|c|} \hline \alpha_i \\ \hline \beta_i \end{array} \{\alpha_i \rightarrow \beta_i\}
\end{aligned} \tag{3.18}$$

3. Derivative acting on field three and contracted with field one and two:

$$\Phi_{\alpha}^{(1)} \Phi^{(2),\dot{\alpha}} \left(D_{\dot{\alpha}}^{\alpha} D^{n_2-1} \Phi^{(3)} \right) \dots \simeq - \Phi_{\alpha}^{(1)} \Phi^{(2),\dot{\alpha}} \left(D^{n_2-1} \Phi^{(3)} \right) \dots (D_{\dot{\alpha}}^{\alpha} \dots), \tag{3.19}$$

This can be rewritten in terms of Young Diagrams:

$$\begin{aligned}
\overline{\begin{array}{|c|} \hline 2 \\ \hline 3 \end{array}} \otimes \begin{array}{|c|} \hline 1 \\ \hline 3 \end{array} \Phi_{\alpha_1}^{(1)} \Phi^{(2),\dot{\alpha}_2} D_{\alpha_3}^{\dot{\alpha}_3} \Phi^{(3)} \dots = - \sum_{k=4}^N \overline{\begin{array}{|c|} \hline 2 \\ \hline k \end{array}} \otimes \begin{array}{|c|} \hline 1 \\ \hline k \end{array} \Phi_{\alpha_1}^{(1)} \Phi^{(2),\dot{\alpha}_2} \Phi^{(3)} \dots D_{\alpha_k}^{\dot{\alpha}_k} \Phi^{(k)} \dots \\
+ \overline{\begin{array}{|c|} \hline \dot{\alpha}_2 \\ \hline \dot{\alpha}_1 \end{array}} \otimes \begin{array}{|c|} \hline \alpha_1 \\ \hline \beta_1 \end{array} D_{\alpha_1}^{\dot{\alpha}_1} \Phi_{\beta_1}^{(1)} \Phi^{(2),\dot{\alpha}_2} \Phi^{(3)} \dots \\
+ \overline{\begin{array}{|c|} \hline \dot{\alpha}_2 \\ \hline \beta_2 \end{array}} \otimes \begin{array}{|c|} \hline \alpha_1 \\ \hline \alpha_2 \end{array} \Phi_{\alpha_1}^{(1)} D_{\alpha_2}^{\dot{\alpha}_2} \Phi^{(2),\dot{\beta}_2} \Phi^{(3)} \dots \\
\Leftrightarrow \overline{\begin{array}{|c|} \hline 2 \\ \hline 3 \end{array}} \otimes \begin{array}{|c|} \hline 1 \\ \hline 3 \end{array} = - \sum_{k=4}^N \overline{\begin{array}{|c|} \hline 2 \\ \hline k \end{array}} \otimes \begin{array}{|c|} \hline 1 \\ \hline k \end{array} + \overline{\begin{array}{|c|} \hline \dot{\alpha}_2 \\ \hline \dot{\alpha}_1 \end{array}} \otimes \begin{array}{|c|} \hline \alpha_1 \\ \hline \beta_1 \end{array} \{\alpha_1 \rightarrow \beta_1\} + \overline{\begin{array}{|c|} \hline \dot{\alpha}_2 \\ \hline \beta_2 \end{array}} \otimes \begin{array}{|c|} \hline \alpha_1 \\ \hline \alpha_2 \end{array} \{\dot{\alpha}_2 \rightarrow \dot{\beta}_2\}
\end{aligned} \tag{3.20}$$

There exists of course the same relation with changed indices:

$$\Phi^{(1),\dot{\alpha}} \Phi_{\alpha}^{(2)} \left(D_{\dot{\alpha}}^{\alpha} D^{n_2-1} \Phi^{(3)} \right) \dots \simeq - \Phi^{(1),\dot{\alpha}} \Phi_{\alpha}^{(2)} \left(D^{n_2-1} \Phi^{(3)} \right) \dots (D_{\dot{\alpha}}^{\alpha} \dots), \tag{3.21}$$

This can be rewritten in terms of Young Diagrams:

$$\begin{aligned}
\overline{\begin{array}{|c|} \hline 1 \\ \hline 3 \end{array}} \otimes \begin{array}{|c|} \hline 2 \\ \hline 3 \end{array} \Phi^{(1),\dot{\alpha}_1} \Phi_{\alpha_2}^{(2)} D_{\alpha_3}^{\dot{\alpha}_3} \Phi^{(3)} \dots = - \sum_{k=4}^N \overline{\begin{array}{|c|} \hline 1 \\ \hline k \end{array}} \otimes \begin{array}{|c|} \hline 2 \\ \hline k \end{array} \Phi^{(1),\dot{\alpha}_1} \Phi_{\alpha_2}^{(2)} \Phi^{(3)} \dots D_{\alpha_k}^{\dot{\alpha}_k} \Phi^{(k)} \dots \\
+ \overline{\begin{array}{|c|} \hline \dot{\alpha}_1 \\ \hline \beta_1 \end{array}} \otimes \begin{array}{|c|} \hline \alpha_2 \\ \hline \alpha_1 \end{array} D_{\alpha_1}^{\dot{\alpha}_1} \Phi^{(1),\dot{\beta}_1} \Phi_{\alpha_2}^{(2)} \Phi^{(3)} \dots \\
+ \overline{\begin{array}{|c|} \hline \dot{\alpha}_1 \\ \hline \dot{\alpha}_2 \end{array}} \otimes \begin{array}{|c|} \hline \alpha_2 \\ \hline \beta_2 \end{array} \Phi^{(1),\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} \Phi_{\beta_2}^{(2)} \Phi^{(3)} \dots \\
\Leftrightarrow \overline{\begin{array}{|c|} \hline 1 \\ \hline 3 \end{array}} \otimes \begin{array}{|c|} \hline 2 \\ \hline 3 \end{array} = - \sum_{k=4}^N \overline{\begin{array}{|c|} \hline 1 \\ \hline k \end{array}} \otimes \begin{array}{|c|} \hline 2 \\ \hline k \end{array} + \overline{\begin{array}{|c|} \hline \dot{\alpha}_1 \\ \hline \beta_1 \end{array}} \otimes \begin{array}{|c|} \hline \alpha_2 \\ \hline \alpha_1 \end{array} \{\dot{\alpha}_1 \rightarrow \dot{\beta}_1\} + \overline{\begin{array}{|c|} \hline \dot{\alpha}_1 \\ \hline \dot{\alpha}_2 \end{array}} \otimes \begin{array}{|c|} \hline \alpha_2 \\ \hline \beta_2 \end{array} \{\alpha_2 \rightarrow \beta_2\}
\end{aligned} \tag{3.22}$$

4. Two derivatives acting on the second and third field are substituted by:

$$\begin{aligned}
\Phi_1 D^2 (\Phi_2 \Phi_3 \dots) &= - D^2 \Phi_1 \Phi_2 \Phi_3 \dots \\
&= \Phi_1 (D^2 \Phi_2) \Phi_3 \dots + \Phi_1 \Phi_2 (D^2 \Phi_3) \dots + \Phi_1 \Phi_2 \Phi_3 (D^2 \dots) \\
&\quad + 2 \Phi_1 (D \Phi_2) (D \Phi_3) \dots + 2 \Phi_1 (D \Phi_2) \Phi_3 (D \dots) + 2 \Phi_1 \Phi_2 (D \Phi_3) (D \dots),
\end{aligned} \tag{3.23}$$

5. Schouten identities have to be implemented manually by substituting

$$\begin{aligned}
\varepsilon_{\alpha_i \alpha_l} \varepsilon_{\alpha_j \alpha_k} &= - \varepsilon_{\alpha_i \alpha_j} \varepsilon_{\alpha_k \alpha_l} - \varepsilon_{\alpha_i \alpha_k} \varepsilon_{\alpha_l \alpha_j} \\
\tilde{\varepsilon}_{\dot{\alpha}_i \dot{\alpha}_l} \tilde{\varepsilon}_{\dot{\alpha}_j \dot{\alpha}_k} &= - \tilde{\varepsilon}_{\dot{\alpha}_i \dot{\alpha}_j} \tilde{\varepsilon}_{\dot{\alpha}_k \dot{\alpha}_l} - \tilde{\varepsilon}_{\dot{\alpha}_i \dot{\alpha}_k} \tilde{\varepsilon}_{\dot{\alpha}_l \dot{\alpha}_j}
\end{aligned} \tag{3.24}$$

Algorithm idea TODO: Update

1. Simplify all anti symmetrised $\text{SL}(2, \mathbb{C})$ indices with the help of (4.3), (4.4) and (4.5).
2. Insert expression for anti symmetrised $\text{SL}(2, \mathbb{C})$ indices and split terms eventually.
3. Apply one iteration of the above mentioned IBP-relations, i.e. only one derivative should be changed.
4. Redo step 1 to 3 and so on, up to the point where all IBP relations are applied.
5. Rewrite ε with Schouten identities.

After this procedure, all initially written EOMs should be removed and also most of the EOMs which only occur via IBP-relations should be gone. Notice, the above procedure essentially changes only derivatives on the first 3 fields.

Applying of all the relations brings operators in the form: $\mathcal{M} = \sum_i m_i \mathcal{M}_i^{(y)}$, where $\mathcal{M}^{(y)}$ consist of fields of the same type only (s. e.g. (3.26)).

The only relation relevant for the above considered examples is the last one and yields for the 35th term, (3.10):

$$\begin{aligned} (T^I T^J)_i^j H^{\dagger, i} H_j V_{\mu\nu}^I V^{J, \mu\nu} = & -\frac{1}{16} \left(2\varepsilon^{\alpha_1 \alpha_2} \varepsilon^{\beta_1 \beta_2} - \varepsilon^{\alpha_1 \beta_1} \varepsilon^{\alpha_2 \beta_2} \right) \varepsilon^{ip} \varepsilon^{pl} \varepsilon^{kn} \varepsilon^{mq} \varepsilon^{qj} H_i H_j V_{L\alpha_1 \beta_1, (kl)} V_{L\alpha_2 \beta_2, (mn)} \\ & -\frac{1}{16} \left(2\varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} \varepsilon_{\dot{\beta}_1 \dot{\beta}_2} - \varepsilon_{\dot{\alpha}_1 \dot{\beta}_1} \varepsilon_{\dot{\alpha}_2 \dot{\beta}_2} \right) \varepsilon^{ip} \varepsilon^{pl} \varepsilon^{kn} \varepsilon^{mq} \varepsilon^{qj} H_i H_j V_{R, (kl)}^{\dot{\alpha}_1 \dot{\beta}_1} V_{R, (mn)}^{\dot{\alpha}_2 \dot{\beta}_2}. \end{aligned} \quad (3.25)$$

Identify the following tensors

$$\begin{aligned} \mathcal{M}_1^{(y)} &= H_i H_j V_{L\alpha_1 \beta_1, (kl)} V_{L\alpha_2 \beta_2, (mn)}, & T_{\text{SU}(2)1}^{(y)} &= \varepsilon^{ip} \varepsilon^{pl} \varepsilon^{kn} \varepsilon^{mq} \varepsilon^{qj}, \\ \mathcal{M}_2^{(y)} &= H_i H_j V_{R, (kl)}^{\dot{\alpha}_1 \dot{\beta}_1} V_{R, (mn)}^{\dot{\alpha}_2 \dot{\beta}_2}, \end{aligned} \quad (3.26)$$

3.4 Gauge groups

First, some group theory stuff [s. 6, S. 482–488]:

$$[T^A, T^B] = i f^{ABC} T^C, \quad \text{Tr}(T^A T^B) = T_{ji}^A T_{ij}^B = T_F \delta^{AB}, \quad \sum_A (T^A T^A)_{ij} = C_F \delta_{ij}, \quad (3.27)$$

$$f^{ACD} f^{BCD} = C_A \delta^{AB}, \quad f^{ABC} = -\frac{i}{T_F} \text{Tr}([T^A, T^B] T^C), \quad (3.28)$$

$$\sum_A T_{ij}^A T_{kl}^A = T_F \left(\delta_{il} \delta_{kj} - \frac{1}{d_F} \delta_{ij} \delta_{kl} \right), \quad (3.29)$$

where $T_F = \frac{1}{2}$, $C_A = N = d_F$ and $C_F = \frac{N^2-1}{2N}$ for $\text{SU}(N)$. $(T^A)^b_c := \frac{(\lambda^A)^b_c}{2}$ for $\text{SU}(3)$ with Gell-Mann matrices λ^A and $(T^I)^j_k := \frac{(\tau^I)^j_k}{2}$ for $\text{SU}(2)$, with Pauli matrices τ^I . Note further that the structure constants f^{ABC} are totally antisymmetric and (3.29) is essentially the Fierz identity for $\text{SU}(N)$.

$$\begin{aligned} \varepsilon_{acd} (\lambda^A)^d_b G^A &= G_{abc} \sim \boxed{\frac{a}{c} \frac{b}{c}} \Leftrightarrow G^A = \frac{(\lambda^A)^b_e}{4} \varepsilon^{ace} G_{abc}, \\ \varepsilon_{abc} (Q^\dagger)^c &= Q_{ab}^\dagger \sim \boxed{\frac{a}{b}} \Leftrightarrow (Q^\dagger)^c = \frac{1}{2} \varepsilon^{abc} Q_{bc}^\dagger, \\ \varepsilon_{jk} (\tau^I)^k_i W^I &= W_{(ij)} \sim \boxed{\frac{i}{j}} \Leftrightarrow W^I = \frac{(\tau^I)^i_j}{2} \varepsilon^{jk} W_{(ik)}, \\ \varepsilon_{ij} (H^\dagger)^j &= H_i \sim \boxed{i} \Leftrightarrow (H^\dagger)^i = \varepsilon^{ij} H_j, \end{aligned} \quad (3.30)$$

where $W_{(ik)} := (W_{ik} + W_{ki})/2$ is the symmetric part of W . Note that also $u_{\mathbb{C}}$ and $d_{\mathbb{C}}$ are transforming in the anti-fundamental representation of $SU(3)$ and are rewritten similar as $Q^\dagger$. Further, also parts of the lepton and quark doublet, $L^\dagger$, $Q^\dagger$ and $e_{\mathbb{C}}$ are rewritten like the Higgs field H .

Gauge projection Express now the gauge group structures which occurred above (e.g. (3.26)) and are part of the singlet representation of the gauge group by basis tensors of the singlet basis for the gauge group we already know. This is essentially simplified by the fact, that it is possible to define a scalar (inner) product between two gauge group tensors T_1 and T_2 :

$$(T_1, T_2) := \sum_{a_1, a_2, \dots} T_1^{a_1 a_2 \dots} T_2^{a_1 a_2 \dots}. \quad (3.31)$$

The singlet basis vectors are denoted by $\{e_i := (T_G^{\tilde{y}})_i\}$ and are not necessarily orthogonal. Therefore, the introduction of the symmetric Gram-matrix $G_{ij} := (e_i, e_j)$ consisting all possible scalar products of the basis tensors is necessary and after defining the tensor $v_i := (G_{ij})^{-1} e_j$ allows to express an arbitrary tensor in the singlet- or y-basis:

$$T_G = \sum_i (v_i, T_G) e_i, \quad (3.32)$$

where the Gram-matrix ensures normalization $(e_i, v_i) = 1$.

Combination of the Lorentz group decomposition relations and projections in the gauge group sector allows to write every operator in the y-basis as

$$\mathcal{O} = T_G \mathcal{M} = \sum_{i=1}^{\mathcal{N}_G} t_i T_G^{\tilde{y}} \sum_{j=1}^{\mathcal{N}_M} m_j \mathcal{M}_j^{(y)} =: \sum_{i,j} t_i m_j \mathcal{O}_{i,j}^{(y)}. \quad (3.33)$$

$\mathcal{N}_G$ and $\mathcal{N}_M$ denote the dimension of the spaces spanned by T_G and $\mathcal{M}$.

3.5 Redundant operators

The operators of Benjamin Summ, now in the correct notation, do not necessarily form a basis, but the operators of the y-basis do. Thus, it seems reasonable to extract an linearly independent set of operators from the BS operators. The procedure can be described as follows:

1. Search for operators (in BS's output) build out of the same field.
2. Express every operator in the y-Basis as described above, so that all operators have the form⁵ $\mathcal{O} = \kappa_i \mathcal{O}_i^{(y)}$.
3. Construct a matrix out of all coefficients:
$$\begin{pmatrix} \mathcal{O} \\ \mathcal{O}' \\ \vdots \end{pmatrix} = \begin{pmatrix} \kappa_1 & \kappa_2 & \dots \\ \kappa'_1 & \kappa'_2 & \dots \\ \vdots & & \ddots \end{pmatrix} \begin{pmatrix} \mathcal{O}_1^{(y)} \\ \mathcal{O}_2^{(y)} \\ \vdots \end{pmatrix} =: K_{ij} \mathcal{O}_i^{(y)}$$
4. Determine rank $r := \text{rank}(K_{ij})$ and choose r linearly independent rows of the matrix K_{ij} and build from them the new matrix $\bar{K}_{ij}$.

This gives the so called m-basis, which is defined by the r linearly independent operators:

$$\mathcal{O}_i^{(m)} = \sum_{j=1}^{\mathcal{N}} \bar{K}_{ij} \mathcal{O}_j^{(y)}, \quad i = 1, \dots, r. \quad (3.34)$$

The operators in the m-basis span the whole space of the y-basis if and only if $r = \mathcal{N}$.

⁵For ease of notation an general index $i := (i, j)$ was introduced, which yield for (3.33): $\mathcal{O} = t_i m_j \mathcal{O}_{i,j}^{(y)} \Leftrightarrow \mathcal{O} = \kappa_i \mathcal{O}_i^{(y)}$

4 Equation of Motion complication

4.1 Antisymmetric $SL(2, \mathbb{C})$ indices

Consider the following relations

$$\begin{aligned}\varepsilon^{\beta\alpha} D_{\alpha}^{\dot{\alpha}} D_{\beta}^{\dot{\beta}} &= D^{\dot{\alpha}\alpha} D_{\alpha}^{\dot{\beta}} \equiv (DD)^{\dot{\alpha}\dot{\beta}} = D_{\mu} D^{\mu} \varepsilon^{\dot{\alpha}\dot{\beta}} - \frac{i}{2} [D_{\mu}, D_{\nu}] (\bar{\sigma}^{\mu\nu})^{\dot{\alpha}\dot{\beta}}, \\ \varepsilon_{\dot{\beta}\dot{\alpha}} D_{\alpha}^{\dot{\alpha}} D_{\beta}^{\dot{\beta}} &= D_{\alpha\dot{\beta}} D_{\beta}^{\dot{\beta}} \equiv (DD)_{\alpha\beta} = D_{\mu} D^{\mu} \varepsilon_{\alpha\beta} - \frac{i}{2} [D_{\mu}, D_{\nu}] (\sigma^{\mu\nu})_{\alpha\beta}.\end{aligned}\tag{4.1}$$

The commutator of covariant derivatives generates (a sum of) field strength tensors

$$i[D_{\mu}, D_{\nu}] \equiv F_{\mu\nu} \equiv B_{\mu\nu} + W_{\mu\nu}^I T_{SU(2)}^I + G_{\mu\nu}^A T_{SU(3)}^A,\tag{4.2}$$

denoted by $F_{\mu\nu}$. The definitions of the left and right handed field strength tensors in eq. 2.8 can then be identified and used in these expressions:

$$\begin{aligned}\varepsilon^{\beta\alpha} D_{\alpha}^{\dot{\alpha}} D_{\beta}^{\dot{\beta}} &= D^{\dot{\alpha}\alpha} D_{\alpha}^{\dot{\beta}} \equiv (DD)^{\dot{\alpha}\dot{\beta}} = D_{\mu} D^{\mu} \varepsilon^{\dot{\alpha}\dot{\beta}} - iF_R^{\dot{\alpha}\dot{\beta}}, \\ \varepsilon_{\dot{\beta}\dot{\alpha}} D_{\alpha}^{\dot{\alpha}} D_{\beta}^{\dot{\beta}} &= D_{\alpha\dot{\beta}} D_{\beta}^{\dot{\beta}} \equiv (DD)_{\alpha\beta} = D_{\mu} D^{\mu} \varepsilon_{\alpha\beta} + iF_{L\alpha\beta}.\end{aligned}\tag{4.3}$$

$$\begin{aligned}\varepsilon^{\alpha\beta} D_{\alpha}^{\dot{\alpha}} \psi_{\beta} &= -(\not{D}\psi_L)^{\dot{\alpha}} \text{ for } \psi_L := \begin{pmatrix} \psi_{\alpha} \\ 0 \end{pmatrix}, \\ \varepsilon_{\dot{\alpha}\dot{\beta}} D_{\alpha}^{\dot{\alpha}} \psi^{\dot{\beta}} &= -(\not{D}\psi_R)_{\alpha} \text{ for } \psi_R := \begin{pmatrix} 0 \\ \psi^{\dot{\beta}} \end{pmatrix}, \\ \varepsilon_{\dot{\alpha}\dot{\beta}} D_{\alpha}^{\dot{\alpha}} \psi^{\dot{\beta}} &= +(\bar{\psi}_L \overleftarrow{\not{D}})_{\alpha} \text{ for } \bar{\psi}_L := (0, \psi_{\dot{\alpha}}^{\dagger}), \\ \varepsilon^{\alpha\beta} D_{\alpha}^{\dot{\alpha}} \psi_{\beta} &= +(\bar{\psi}_R \overleftarrow{\not{D}})^{\dot{\alpha}} \text{ for } \bar{\psi}_R := (\psi_{\alpha}, 0),\end{aligned}\tag{4.4}$$

$$\begin{aligned}\varepsilon_{\dot{\alpha}\dot{\beta}} D_{\gamma}^{\dot{\alpha}} (B_R)^{\dot{\beta}\dot{\gamma}} &= D_{\mu} B^{\mu\nu} (\bar{\sigma}_{\nu})^{\dot{\gamma}}_{\dot{\gamma}}, \\ \varepsilon^{\alpha\beta} D_{\alpha}^{\dot{\alpha}} (B_L)_{\beta\dot{\gamma}} &= D_{\mu} B^{\mu\nu} (\bar{\sigma}_{\nu})^{\dot{\gamma}}_{\dot{\gamma}},\end{aligned}\tag{4.5}$$

which are equivalently ⁶ to the following

$$\boxed{\begin{aligned}D_{[\alpha}^{\dot{\alpha}} D_{\beta]}^{\dot{\beta}} &= \frac{1}{2} \varepsilon_{\alpha\beta} \varepsilon^{\dot{\alpha}\dot{\beta}} D_{\mu} D^{\mu} - \frac{i}{2} \varepsilon_{\alpha\beta} F_R^{\dot{\alpha}\dot{\beta}}, \\ D_{[\alpha}^{\dot{\alpha}} \psi_{\beta]} &= \frac{1}{2} \varepsilon_{\alpha\beta} (\not{D}\psi_L)^{\dot{\alpha}}, \\ D_{\gamma}^{[\dot{\alpha}} (B_R)^{\dot{\beta}]\dot{\gamma}} &= -\frac{1}{2} \varepsilon^{\dot{\alpha}\dot{\beta}} D_{\mu} B^{\mu\nu} (\bar{\sigma}_{\nu})^{\dot{\gamma}}_{\dot{\gamma}}, \\ D_{[\alpha}^{\dot{\alpha}} (B_L)_{\beta]\dot{\gamma}} &= -\frac{1}{2} \varepsilon_{\alpha\beta} D_{\mu} B^{\mu\nu} (\bar{\sigma}_{\nu})^{\dot{\gamma}}_{\dot{\gamma}}.\end{aligned}}\tag{4.6}$$

For the occurring equation of motion (EOM)

$$\begin{aligned}D^2 \phi &= J_{\phi}, \quad i\not{D}\psi = J_{\psi} \\ D^{\mu} B_{\mu\nu} &= J_{U(1),\nu}, \quad D^{\mu} W_{\mu\nu}^I = J_{SU(2),\nu}^I, \quad D^{\mu} G_{\mu\nu}^A = J_{SU(3),\nu}^A,\end{aligned}\tag{4.7}$$

⁶Consider two $SL(2, \mathbb{C})$ -tensors $S_{\alpha\beta}$, $T^{\dot{\alpha}\dot{\beta}}$ and define $S := \text{Tr}(S) := S_{\alpha}^{\alpha}$, $T := \text{Tr}(T) := T^{\dot{\alpha}}_{\dot{\alpha}}$. Each tensor can be decomposed in the form $S_{\alpha\beta} = S_{(\alpha\beta)} + S_{[\alpha\beta]}$, with $S_{(\alpha\beta)} := (S_{\alpha\beta} + S_{\beta\alpha})/2$ and $S_{[\alpha\beta]} := (S_{\alpha\beta} - S_{\beta\alpha})/2$. The antisymmetric part of both tensor is the given by an epsilon tensor:

$$S_{[\alpha\beta]} = -\frac{1}{2} \varepsilon_{\alpha\beta} S = -\frac{1}{2} \varepsilon_{\alpha\beta} \varepsilon^{\gamma\delta} S_{\gamma\delta}, \quad T^{[\dot{\alpha}\dot{\beta}]} = -\frac{1}{2} \varepsilon^{\dot{\alpha}\dot{\beta}} T = -\frac{1}{2} \varepsilon^{\dot{\alpha}\dot{\beta}} \varepsilon_{\dot{\gamma}\dot{\delta}} T^{\dot{\gamma}\dot{\delta}}.$$

the kinetic term should be substituted by the source term, since operators differing only by EOM are physically equivalent.

These relations are to be replaced until all spinor indices of the remaining part are completely symmetrical. The further terms generated in the process should be further translated separately.

$$\begin{aligned}\mathcal{L}_{\text{SMEFT}} &= \tilde{\mathcal{C}}_n^{(4)} \tilde{\mathcal{O}}_n^{(4)} + \frac{1}{\Lambda} \tilde{\mathcal{C}}_n^{(5)} \tilde{\mathcal{O}}_n^{(5)} + \mathcal{O}\left(\frac{1}{\Lambda^2}\right) \\ i\tilde{\not{D}}\psi &= \Lambda \tilde{J}^{(4)}(\tilde{\mathcal{C}}_n^{(4)}) + \tilde{J}^{(5)}(\tilde{\mathcal{C}}_n^{(5)}) + \mathcal{O}\left(\frac{1}{\Lambda}\right)\end{aligned}\tag{4.8}$$

4.2 Substitution of EOMs via Field redefinitions

This section explains the substitution of equation of motion operators in an effective field theory, due to a field redefinition [s. 8, p. 337–338]. The effective field theory Lagrangian (s. eq. 1.1) might also be rewritten in form of the Action

$$S[\phi_i] = S_0[\phi_i] + \frac{1}{\Lambda} S_1[\phi_i] + \frac{1}{\Lambda^2} S_2[\phi_i] + \mathcal{O}\left(\frac{1}{\Lambda^3}\right).\tag{4.9}$$

We assume now that there exists a term $S_n^{\text{EOM}}[\phi_i]$ in this expansion which would when ϕ_i is chosen such that it obeys the equation of motion. This would look for example like

$$S_n^{\text{EOM}}[\phi_i] = \int d^4x f^i(\phi_i, \partial_\mu \phi_i, x) \frac{\delta S}{\delta \phi_i}.\tag{4.10}$$

If we then consider the following transformation of the field

$$\phi_i(x) \rightarrow \phi'_i(x) := \phi_i(x) - \frac{1}{\Lambda^n} f^i(\phi_i, \partial_\mu \phi_i, x),\tag{4.11}$$

the Action transforms in such a way that the equation of motion term is exactly cancelled:

$$S[\phi_i] \rightarrow S[\phi_i - \frac{1}{\Lambda^n} f^i] = S[\phi_i] - \frac{1}{\Lambda^n} \int d^4x f^i(\phi_i, \partial_\mu \phi_i, x) \frac{\delta S}{\delta \phi_i} + \mathcal{O}\left(\frac{1}{\Lambda^{2n}}\right).\tag{4.12}$$

We stress that, first, the interaction term $S_n^{\text{EOM}}[\phi_i]$ can be removed by such a field redefinition **without** effecting any other terms of lower or equal order in the mass scale Λ . Second, the neglected higher order terms are usually of quadratic order in the scale considered for the term f^i , due to the quadratic structure of the kinetic term. Finally, it is **not necessary** to even know which equation of motion the field obeys exactly, since the field redefinition which is needed for the cancellation only demands the prefactor of the equation of motion term.

Example Consider the term $\frac{1}{\Lambda^2} \phi^3 \square \phi$ with a real scalar field ϕ . The transformation to get rid of such an equation of motion term substitute $\phi \rightarrow \phi - \frac{1}{\Lambda^2} \phi^3$:

$$\begin{aligned}\frac{1}{2} \phi \square \phi + \frac{1}{\Lambda^2} \phi^3 \square \phi &\rightarrow \frac{1}{2} (\phi - \frac{1}{\Lambda^2} \phi^3) \square (\phi - \frac{1}{\Lambda^2} \phi^3) + \frac{1}{\Lambda^2} \phi^3 \square \phi + \mathcal{O}\left(\frac{1}{\Lambda^4}\right) \\ &= \frac{1}{2} \phi \square \phi - \frac{1}{2\Lambda^2} \phi^3 \square \phi - \frac{1}{2\Lambda^2} \underbrace{\phi \square \phi^3}_{\stackrel{7}{=} \phi^3 \square \phi} + \frac{1}{\Lambda^2} \phi^3 \square \phi + \mathcal{O}\left(\frac{1}{\Lambda^4}\right) \\ &= \frac{1}{2} \phi \square \phi + \mathcal{O}\left(\frac{1}{\Lambda^4}\right).\end{aligned}\tag{4.13}$$

Nevertheless, also terms without a derivative are transformed. For example:

⁷Note: $\frac{1}{4} \square \phi^4 = \partial_\mu (\phi^3 \partial^\mu \phi) = 3\phi^2 \partial_\mu \phi \partial^\mu \phi + \phi \square \phi^3$
 $\Rightarrow \phi \square \phi^3 = 3\phi \partial_\mu (\phi^2 \partial^\mu \phi) = 6\phi^2 \partial_\mu \phi \partial^\mu \phi + 3\phi^3 \square \phi = -2\phi^3 \square \phi + \partial_\mu (\dots) + 3\phi^3 \square \phi = \phi^3 \square \phi$

- $m^2\phi^2 \rightarrow m^2\phi^2 - 2\frac{m^2}{\Lambda^2}\phi^4 + \mathcal{O}(\Lambda^{-4})$
- $\lambda\phi^4 \rightarrow \lambda\phi^4 - 4\frac{\lambda}{\Lambda^2}\phi^6 + \mathcal{O}(\Lambda^{-4})$
- $\frac{\mathcal{C}_6}{\Lambda^2}\phi^6 \rightarrow \frac{\mathcal{C}_6}{\Lambda^2}\phi^6 + \mathcal{O}(\Lambda^{-4})$

$$\Rightarrow \begin{cases} \lambda & \rightarrow \lambda - 2\frac{m^2}{\Lambda^2}, \\ \mathcal{C}_6 & \rightarrow \mathcal{C}_6 - 4\lambda \end{cases}.$$

5 Program

TODO: Structure of the program.

Appendix

A Notation

A.1 Lorentz and $\text{SL}(2, \mathbb{C})$ invariant tensors

σ^μ and $\bar{\sigma}^\mu$ are defined with⁸

$$\boxed{(\sigma^\mu)_{\alpha\dot{\alpha}} := (\mathbb{1}_{\alpha\dot{\alpha}}, (\sigma^i)_{\alpha\dot{\alpha}})^\mu \quad \wedge \quad (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} := (\mathbb{1}^{\dot{\alpha}\alpha}, -(\sigma^i)^{\dot{\alpha}\alpha})^\mu,} \quad (\text{A.1})$$

and are hermitian and traceless. Further, they obey the properties:

$$\bar{\sigma}^\mu \sigma^\nu + \bar{\sigma}^\nu \sigma^\mu = 2g^{\mu\nu} \mathbb{1}, \quad \sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu = 2g^{\mu\nu} \mathbb{1}, \quad (\text{A.2})$$

$$\text{Tr}(\bar{\sigma}^\mu \sigma^\nu) = \text{Tr}(\sigma^\mu \bar{\sigma}^\nu) \equiv \varepsilon^{\alpha\beta} \varepsilon^{\dot{\alpha}\dot{\beta}} (\sigma^\mu)_{\alpha\dot{\alpha}} (\sigma^\nu)_{\beta\dot{\beta}} = 2g^{\mu\nu} \quad (\text{A.3})$$

$$\boxed{(\sigma^\mu)^{\alpha\dot{\alpha}} (\sigma_\mu)^{\beta\dot{\beta}} = 2\varepsilon^{\alpha\beta} \varepsilon^{\dot{\alpha}\dot{\beta}}.} \quad (\text{A.4})$$

Dotted and undotted indices can be raised and lowered by "metric"-tensor

$$\begin{aligned} \varepsilon^{\alpha\beta} &:= (\varepsilon)^{\alpha\beta} = -\varepsilon^{\beta\alpha} \quad \wedge \quad \varepsilon^{\dot{\alpha}\dot{\beta}} := (\varepsilon)^{\dot{\alpha}\dot{\beta}}, \\ \varepsilon_{\alpha\beta} &:= (\varepsilon^{-1})_{\alpha\beta} \quad \wedge \quad \varepsilon_{\dot{\alpha}\dot{\beta}} := (\varepsilon^{-1})_{\dot{\alpha}\dot{\beta}}, \\ \delta_\beta^\alpha &= \varepsilon^{\alpha\gamma} \varepsilon_{\gamma\beta} \quad \wedge \quad \delta_{\dot{\beta}}^{\dot{\alpha}} = \varepsilon^{\dot{\alpha}\dot{\gamma}} \varepsilon_{\dot{\gamma}\dot{\beta}}, \end{aligned} \quad (\text{A.5})$$

which is invariant w.r.t $\text{SL}(2, \mathbb{C})$ -transformation and defined by the matrix⁹

$$\varepsilon := +i\sigma^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \varepsilon = \varepsilon^* = -\varepsilon^\dagger = -\varepsilon^T = -\varepsilon^{-1}. \quad (\text{A.6})$$

As can be checked by considering the explicit transformations (2.2), the following expressions transform in the correct way and establish thus the index raising and lowering:

$$\xi^\alpha = \varepsilon^{\alpha\beta} \xi_\beta, \quad \xi_\alpha = \varepsilon_{\alpha\beta} \xi^\beta, \quad \bar{\chi}^{\dot{\alpha}} = \varepsilon^{\dot{\alpha}\dot{\beta}} \bar{\chi}_{\dot{\beta}}, \quad \bar{\chi}_{\dot{\alpha}} = \varepsilon_{\dot{\alpha}\dot{\beta}} \bar{\chi}^{\dot{\beta}}. \quad (\text{A.7})$$

Further invariant contractions of spinors are defined by

$$\begin{aligned} \xi_1 \xi_2 &:= (\xi_1)^\alpha (\xi_2)_\alpha = (\xi_1)_2 (\xi_2)_1 - (\xi_1)_1 (\xi_2)_2 = -(\xi_1)_\alpha (\xi_2)^\alpha, \\ \bar{\chi}_1 \bar{\chi}_2 &:= (\bar{\chi}_1)_{\dot{\alpha}} (\bar{\chi}_2)^{\dot{\alpha}} = (\bar{\chi}_1)^{\dot{1}} (\bar{\chi}_2)^{\dot{2}} - (\bar{\chi}_1)^{\dot{2}} (\bar{\chi}_2)^{\dot{1}}. \end{aligned} \quad (\text{A.8})$$

This definition automatically respects anti-commutation of fermions, e.g. $\xi\phi = \xi^\alpha \phi_\alpha = -\phi_\alpha \xi^\alpha = \phi^\alpha \xi_\alpha = \phi\xi$.

In addition,

$$\boxed{(\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} = \varepsilon^{\alpha\beta} \varepsilon^{\dot{\alpha}\dot{\beta}} (\sigma^\mu)_{\beta\dot{\beta}} = (\sigma^\mu)^{\alpha\dot{\alpha}}} \quad (\text{A.9})$$

can be shown. Define now:

$$(\sigma^{\mu\nu})_\alpha{}^\beta := \frac{i}{2} (\sigma^\mu \bar{\sigma}^\nu - \sigma^\nu \bar{\sigma}^\mu)_\alpha{}^\beta, \quad (\bar{\sigma}^{\mu\nu})^{\dot{\alpha}}{}_{\dot{\beta}} := \frac{i}{2} (\bar{\sigma}^\mu \sigma^\nu - \bar{\sigma}^\nu \sigma^\mu)^{\dot{\alpha}}{}_{\dot{\beta}} \quad (\text{A.10})$$

⁸The Pauli matrices are defined as: $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, with $(\sigma^i)^\dagger = \sigma^i$ and $\sigma^i \sigma^j = \delta^{ij} + i\varepsilon^{ijk} \sigma^k$.

⁹Our convention $\varepsilon^{12} = +1 = -\varepsilon_{12}$ is the same as in [5, 9]. A change of sign would especially alter the signs in the charge conjugation matrix in eq. A.25.

and pay attention to the unusual index structure which yields for example:

$$(\sigma^{\mu\nu})^{\alpha\beta} := \frac{i}{2} \varepsilon^{\alpha\gamma} ((\sigma^\mu)_{\gamma\dot{\gamma}} (\bar{\sigma}^\nu)^{\dot{\gamma}\beta} - (\sigma^\nu)_{\gamma\dot{\gamma}} (\bar{\sigma}^\mu)^{\dot{\gamma}\beta}) \quad (\text{A.11})$$

These matrices have the following properties:

$$\sigma^{\mu\nu} = -\sigma^{\nu\mu}, \quad \bar{\sigma}^{\mu\nu} = -\bar{\sigma}^{\nu\mu}, \quad \bar{\sigma}^{\mu\nu} = (\sigma^{\mu\nu})^\dagger, \quad \text{Tr}(\sigma^{\mu\nu}) = \text{Tr}(\bar{\sigma}^{\mu\nu}) = 0, \quad (\text{A.12})$$

$$\sigma^\mu \bar{\sigma}^\nu = g^{\mu\nu} \mathbb{1} - i \sigma^{\mu\nu}, \quad \bar{\sigma}^\mu \sigma^\nu = g^{\mu\nu} \mathbb{1} - i \bar{\sigma}^{\mu\nu}, \quad (\text{A.13})$$

$$\sigma^{0i} = -i \sigma^i, \quad \sigma^{ij} = \epsilon^{ijk} \sigma^k, \quad \bar{\sigma}^{0i} = i \sigma^i, \quad \bar{\sigma}^{ij} = \epsilon^{ijk} \sigma^k, \quad (\text{A.14})$$

$$(\sigma^{\mu\nu})_{\alpha\beta} = (\sigma^{\mu\nu})_{\beta\alpha}, \quad (\bar{\sigma}^{\mu\nu})_{\dot{\alpha}\dot{\beta}} = (\bar{\sigma}^{\mu\nu})_{\dot{\beta}\dot{\alpha}}, \quad (\text{A.15})$$

$$\boxed{(\sigma^{\mu\nu})^{\alpha_1\alpha_2} (\sigma_{\mu\nu})^{\alpha_3\alpha_4} = -4(\varepsilon^{\alpha_1\alpha_4} \varepsilon^{\alpha_2\alpha_3} + \varepsilon^{\alpha_1\alpha_3} \varepsilon^{\alpha_2\alpha_4})} \quad (\text{A.16})$$

$$\boxed{(\sigma^{\mu\nu})_{\dot{\alpha}_1\dot{\alpha}_2} (\sigma_{\mu\nu})_{\dot{\alpha}_3\dot{\alpha}_4} = -4(\varepsilon_{\dot{\alpha}_1\dot{\alpha}_4} \varepsilon_{\dot{\alpha}_2\dot{\alpha}_3} + \varepsilon_{\dot{\alpha}_1\dot{\alpha}_3} \varepsilon_{\dot{\alpha}_2\dot{\alpha}_4})}$$

$$\boxed{(\sigma^{\mu\nu})^{\alpha\beta} (\bar{\sigma}_{\mu\nu})_{\alpha\dot{\beta}} = 0,} \quad (\text{A.17})$$

$$\begin{aligned} \frac{1}{2} \text{Tr}(\sigma^{\mu\nu} \sigma_{\rho\sigma}) &\equiv \frac{1}{2} (\sigma^{\mu\nu})_\alpha{}^\beta (\sigma_{\rho\sigma})_\beta{}^\alpha = \delta_\rho^\mu \delta_\sigma^\nu - \delta_\sigma^\mu \delta_\rho^\nu - i \varepsilon^{\mu\nu}{}_{\rho\sigma}, \\ \frac{1}{2} \text{Tr}(\bar{\sigma}^{\mu\nu} \bar{\sigma}_{\rho\sigma}) &\equiv \frac{1}{2} (\bar{\sigma}^{\mu\nu})^{\dot{\alpha}}{}_{\dot{\beta}} (\bar{\sigma}_{\rho\sigma})^{\dot{\beta}}{}_{\dot{\alpha}} = \delta_\rho^\mu \delta_\sigma^\nu - \delta_\sigma^\mu \delta_\rho^\nu + i \varepsilon^{\mu\nu}{}_{\rho\sigma}. \end{aligned} \quad (\text{A.18})$$

¹⁰ ¹¹and are further self- and anti-self dual, i.e.:

$$\sigma^{\mu\nu} = \frac{i}{2} \varepsilon^{\mu\nu\rho\sigma} \sigma_{\rho\sigma}, \quad \bar{\sigma}^{\mu\nu} = -\frac{i}{2} \varepsilon^{\mu\nu\rho\sigma} \bar{\sigma}_{\rho\sigma}. \quad (\text{A.19})$$

Not further that $\sigma^{\mu\nu}/2$ and $\bar{\sigma}^{\mu\nu}/2$ are generators of the (projective) representations of $L_+^\uparrow$:

$$\begin{aligned} \xi_\alpha(\mathbf{p}) &\rightarrow \xi'_\alpha(\mathbf{p}) = \left(e^{-\frac{i}{2} \omega_{\mu\nu} \frac{\sigma^{\mu\nu}}{2}} \right)_\alpha{}^\beta \xi_\beta(\Lambda^{-1} \mathbf{p}), \\ \bar{\chi}^{\dot{\alpha}}(\mathbf{p}) &\rightarrow \bar{\chi}'^{\dot{\alpha}}(\mathbf{p}) = \left(e^{-\frac{i}{2} \omega_{\mu\nu} \frac{\bar{\sigma}^{\mu\nu}}{2}} \right)^{\dot{\alpha}}{}_{\dot{\beta}} \bar{\chi}^{\dot{\beta}}(\Lambda^{-1} \mathbf{p}). \end{aligned} \quad (\text{A.20})$$

¹⁰See [9, p. 59] or [10, p. 17–19] for a derivation of eq. A.18.

¹¹The four component epsilon tensor $\varepsilon^{\mu\nu\rho\sigma}$ is chosen to fulfil $\varepsilon^{0123} = -\varepsilon_{0123} = +1$.

A.2 Dirac bi-spinor field

Let ξ_α and χ_α be two component left-handed Weyl spinors (i.e. $\xi_\alpha, \chi_\alpha \in (\frac{1}{2}, 0)$). Then, the hermitian conjugate $(\chi^\alpha)^\dagger = \chi^{\dot{\alpha}}$ after raising an index transforms as a right-handed Weyl spinor (i.e. $\chi^{\dot{\alpha}} \in (0, \frac{1}{2})$). Combination of these independent spinors yields the four component Dirac bi-spinor:

$$\psi(x) := \begin{pmatrix} \xi_\alpha(x) \\ (\chi^\dagger)^{\dot{\alpha}}(x) \end{pmatrix} \in \left(\frac{1}{2}, \frac{1}{2}\right). \quad (\text{A.21})$$

Define now the Dirac matrices in the Weyl- or Chiral-representation:

$$\gamma^\mu := \begin{pmatrix} \mathbb{0}_{2 \times 2} & (\sigma^\mu)_{\alpha\dot{\beta}} \\ (\bar{\sigma}^\mu)^{\dot{\alpha}\beta} & \mathbb{0}_{2 \times 2} \end{pmatrix}, \quad \{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}. \quad (\text{A.22})$$

Define also the adjoint Dirac-spinor $\bar{\psi}$:

$$\bar{\psi}(x) := (\chi^\alpha(x), (\xi^\dagger)_{\dot{\alpha}}(x)) = \psi^\dagger(x)\beta, \quad (\text{A.23})$$

where $\beta := \begin{pmatrix} 0 & \delta_{\dot{\beta}}^{\dot{\alpha}} \\ \delta_{\alpha}^{\beta} & 0 \end{pmatrix}$, which is numerically identical to γ^0 , but with a slightly different index structure. Lets define also the charge conjugated field

$$\psi^C(x) := \begin{pmatrix} \chi_\alpha(x) \\ (\xi^\dagger)^{\dot{\alpha}}(x) \end{pmatrix} = C(\bar{\psi})^T(x) \Rightarrow \bar{\psi}^C(x) = (\xi^\alpha(x) \quad (\chi^\dagger)_{\dot{\alpha}}(x)) = -\psi^T(x)C^{-1}, \quad (\text{A.24})$$

with the charge conjugation matrix

$$C := i\gamma^0\gamma^2 = \begin{pmatrix} \varepsilon_{\alpha\beta} & 0 \\ 0 & \varepsilon^{\dot{\alpha}\dot{\beta}} \end{pmatrix} = \begin{pmatrix} -\varepsilon^{\alpha\beta} & 0 \\ 0 & -\varepsilon_{\dot{\alpha}\dot{\beta}} \end{pmatrix}, \quad (\text{A.25})$$

$C = C^* = -C^\dagger = -C^T = -C^{-1}$ and $C^{-1}\gamma_\mu C = -\gamma_\mu^T$. Finally, define

$$\sigma^{\mu\nu} := \frac{i}{2} [\gamma^\mu, \gamma^\nu] = \begin{pmatrix} (\sigma^{\mu\nu})_{\alpha}{}^{\beta} & 0 \\ 0 & (\bar{\sigma}^{\mu\nu})^{\dot{\alpha}}{}_{\dot{\beta}} \end{pmatrix}. \quad (\text{A.26})$$

Combining all the previous results yields the following Dirac bi-spinor expressions:

$$\begin{aligned} \bar{\Psi}_1 \Psi_2 &= \chi_1^\alpha \xi_{2\alpha} + \xi_{1\dot{\alpha}}^\dagger \chi_2^{\dot{\alpha}}, \\ \bar{\Psi}_1 \gamma^\mu \Psi_2 &= \chi_1^\alpha (\sigma^\mu)_{\alpha\dot{\alpha}} \chi_2^{\dot{\alpha}} + \xi_{1\dot{\alpha}}^\dagger (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} \xi_{2\alpha}, \\ \bar{\Psi}_1 \sigma^{\mu\nu} \Psi_2 &= \chi_1^\alpha (\sigma^{\mu\nu})_{\alpha}{}^{\beta} \xi_{2\beta} + \xi_{1\dot{\alpha}}^\dagger (\bar{\sigma}^{\mu\nu})^{\dot{\alpha}}{}_{\dot{\beta}} \chi_2^{\dot{\beta}}, \\ \Psi_1^T C \Psi_2 &= \bar{\Psi}_1^C \Psi_2 = \xi_1^\alpha \xi_{2\alpha} + \chi_{1\dot{\alpha}}^\dagger \chi_2^{\dot{\alpha}}, \\ \Psi_1^T C \gamma^\mu \Psi_2 &= \xi_1^\alpha (\sigma^\mu)_{\alpha\dot{\alpha}} \chi_2^{\dot{\alpha}} + \chi_{1\dot{\alpha}}^\dagger (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} \xi_{2\alpha}, \\ \Psi_1^T C \sigma^{\mu\nu} \Psi_2 &= \xi_1^\alpha (\sigma^{\mu\nu})_{\alpha}{}^{\beta} \xi_{2\beta} + \chi_{1\dot{\alpha}}^\dagger (\bar{\sigma}^{\mu\nu})^{\dot{\alpha}}{}_{\dot{\beta}} \chi_2^{\dot{\beta}}, \\ \bar{\Psi}_1 C \bar{\Psi}_2^T &= \xi_{1\dot{\alpha}}^\dagger \xi_2^{\dot{\alpha}} + \chi_1^\alpha \chi_{2\alpha}, \\ \bar{\Psi}_1 \gamma^\mu C \bar{\Psi}_2^T &= \chi_1^\alpha (\sigma^\mu)_{\alpha\dot{\alpha}} \xi_2^{\dot{\alpha}} + \xi_{1\dot{\alpha}}^\dagger (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} \chi_{2\alpha}, \\ \bar{\Psi}_1 \sigma^{\mu\nu} C \bar{\Psi}_2^T &= \xi_{1\dot{\alpha}}^\dagger (\bar{\sigma}^{\mu\nu})^{\dot{\alpha}}{}_{\dot{\beta}} \xi_2^{\dot{\beta}} + \chi_1^\alpha (\sigma^{\mu\nu})_{\alpha}{}^{\beta} \chi_{2\beta}. \end{aligned} \quad (\text{A.27})$$

Considering the relations between the four component Dirac and the two component Weyl spinors given in eq. 2.7, the above relations can be illustrated with some examples:

- $\bar{u}_R \gamma^\mu u_R = u_C \sigma^\mu u_C^\dagger$ with $u_C^\alpha \equiv \chi^\alpha$ and $(u_C^\dagger)^\alpha \equiv (\chi^\dagger)^\alpha$,
- $\bar{e}_R l_L =: \bar{e} l = e_C L$,
- $\bar{u}^T C d_R =: \bar{u}^C d = u_C^\dagger d_C^\dagger$.

In the output of BSUOLEA and parts of the program the fields given in eq. 2.7 are notated as:

$$\begin{aligned}
 q_L = \mathbf{q} &= \begin{pmatrix} \mathbf{Q} \\ 0 \end{pmatrix}, & u_R = \mathbf{u} &= \begin{pmatrix} 0 \\ [\mathbf{u}_C] \end{pmatrix}, & d_R = \mathbf{d} &= \begin{pmatrix} 0 \\ [\mathbf{d}_C] \end{pmatrix}, \\
 \bar{q}_L = [\mathbf{qbar}] &= (0, [\mathbf{Q}]), & \bar{u}_R = [\mathbf{ubar}] &= ([\mathbf{u}_C], 0), & \bar{d}_R = [\mathbf{dbar}] &= ([\mathbf{d}_C], 0), \\
 l_L = \mathbf{l} &= \begin{pmatrix} \mathbf{L} \\ 0 \end{pmatrix}, & e_R = \mathbf{e} &= \begin{pmatrix} 0 \\ [\mathbf{e}_C] \end{pmatrix}, \\
 \bar{l}_L = [\mathbf{lbar}] &= (0, [\mathbf{L}]), & \bar{e}_R = [\mathbf{ebar}] &= ([\mathbf{e}_C], 0).
 \end{aligned} \tag{A.28}$$

A.3 Fieldstrength tensor

Consider now the chiral basis of gauge bosons $F_{L/R}$ and its relation to the hermitian field F and dual field strength tensor $\tilde{F}$, defined as $\tilde{F}^{\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$:

$$\begin{aligned}
 F_{L/R}^{\mu\nu} &:= \frac{1}{2} (F^{\mu\nu} \pm i \tilde{F}^{\mu\nu}), \quad (F_L)^\dagger = F_R, \\
 \Leftrightarrow F^{\mu\nu} &= F_L^{\mu\nu} + F_R^{\mu\nu}, \quad \tilde{F}^{\mu\nu} = i(F_R^{\mu\nu} - F_L^{\mu\nu}).
 \end{aligned} \tag{A.29}$$

The duality between F and $\tilde{F}$ is altered to the (anti-)self-duality of $F_{L/R}$, i.e. $F_{L/R}^{\mu\nu} = \pm \frac{i}{2} \varepsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}^{L/R}$ which is crucial for the decomposition of the tensor in its irreducible Lorentz representations. Since the (anti-)self-duality condition is Lorentz invariant, both F_L and F_R form irreducible representations of the Lorentz group.

Advantage of this is taken and the left- and right-handed part of the field strength tensor is contracted with the invariant tensors in eq. 2.10:

$$\begin{aligned}
 F_L^{\mu\nu} (\sigma_{\mu\nu})_\alpha^\beta &= F^{\mu\nu} (\sigma_{\mu\nu})_\alpha^\beta = i \tilde{F}^{\mu\nu} (\sigma_{\mu\nu})_\alpha^\beta, & F_L^{\mu\nu} (\bar{\sigma}_{\mu\nu})_{\dot{\alpha}}^{\dot{\beta}} &= 0, \\
 F_R^{\mu\nu} (\bar{\sigma}_{\mu\nu})_{\dot{\alpha}}^{\dot{\beta}} &= F^{\mu\nu} (\bar{\sigma}_{\mu\nu})_{\dot{\alpha}}^{\dot{\beta}} = -i \tilde{F}^{\mu\nu} (\bar{\sigma}_{\mu\nu})_{\dot{\alpha}}^{\dot{\beta}}, & F_R^{\mu\nu} (\sigma_{\mu\nu})_\alpha^\beta &= 0,
 \end{aligned} \tag{A.30}$$

Thus, it seems reasonable to define

$$\boxed{F_{L\beta\alpha} = F_{L\alpha\beta} := \frac{i}{2} F_{\mu\nu} \sigma_{\alpha\beta}^{\mu\nu} \in (1, 0) \text{ and } F_{R\dot{\beta}\dot{\alpha}} = F_{R\dot{\alpha}\dot{\beta}} := -\frac{i}{2} F_{\mu\nu} \bar{\sigma}_{\dot{\alpha}\dot{\beta}}^{\mu\nu} \in (0, 1)} \tag{A.31}$$

where hermitian conjugation changes the chirality, i.e. $(F_{L\alpha\beta})^\dagger = F_{R\dot{\alpha}\dot{\beta}}$. Due to eq. A.18 the inverse is given by

$$\boxed{
 \begin{aligned}
 F_{\mu\nu} &= -\frac{i}{4} \left(F_R^{\dot{\alpha}\dot{\beta}} (\bar{\sigma}_{\mu\nu})_{\dot{\alpha}\dot{\beta}} - F_{L\alpha\beta} \sigma_{\mu\nu}^{\alpha\beta} \right), \\
 \tilde{F}_{\mu\nu} &= -\frac{1}{4} \left(F_R^{\dot{\alpha}\dot{\beta}} (\bar{\sigma}_{\mu\nu})_{\dot{\alpha}\dot{\beta}} + F_{L\alpha\beta} \sigma_{\mu\nu}^{\alpha\beta} \right).
 \end{aligned}
 } \tag{A.32}$$

A.4 Gauge groups

TODO

A.5 (Covariant) Derivative

The covariant derivative is also written in $\text{SL}(2, \mathbb{C})$ -notation

$$D_{\alpha\dot{\alpha}} := D_{\mu}\sigma_{\alpha\dot{\alpha}}^{\mu} \in \left(\frac{1}{2}, \frac{1}{2}\right) \quad (\text{A.33})$$

and since $D_{\dot{\alpha}\alpha} := D_{\mu}\bar{\sigma}_{\dot{\alpha}\alpha}^{\mu} \stackrel{(2.5)}{=} D_{\mu}\sigma_{\alpha\dot{\alpha}}^{\mu} = D_{\alpha\dot{\alpha}}$ it is symmetric in its $\text{SL}(2, \mathbb{C})$ -indices. Index and derivative structure has to be understood as:

$$D\psi_{\alpha\beta\dot{\beta}}\psi_{\gamma} = (D\psi_{\alpha})_{\beta\dot{\beta}}\psi_{\gamma}. \quad (\text{A.34})$$

The derivative D_{μ} is thus equivalent to the $\text{SL}(2, \mathbb{C})$ -expressions:

$$D_{\mu} = \frac{1}{2}D_{\alpha\dot{\alpha}}(\bar{\sigma}_{\mu})^{\dot{\alpha}\alpha} = \frac{1}{2}D_{\dot{\alpha}\alpha}(\sigma_{\mu})^{\alpha\dot{\alpha}} = -\frac{1}{2}D_{\dot{\alpha}}^{\alpha}(\sigma_{\mu})_{\alpha}^{\dot{\alpha}}. \quad (\text{A.35})$$

For Example:

$$D_{\mu}D^{\mu} = \frac{1}{2}D^{\dot{\alpha}\alpha}D_{\alpha\dot{\alpha}}. \quad (\text{A.36})$$

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